

When Does Conformer Geometry Help? Selective Complementarity of 3D Ensemble Statistics and 2D Fingerprints

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Abstract

When do three-dimensional conformer ensembles improve molecular property prediction beyond two-dimensional fingerprints? Despite growing interest in 3D molecular representations, this question lacks a systematic, mechanistically grounded answer. We address it through 13 model configurations across 14 regression targets spanning MoleculeNet, QM9, and MARCEL benchmarks. Our central finding is *selective complementarity*: conformer ensemble statistics—extracted via Distribution Kernel Operators (DKOs)—provide statistically significant gains on solvation-dependent properties (11–14% RMSE reduction, $p < 10^{-4}$, 10-seed validation) but not electronic or steric tasks. This selectivity has a physical basis: the improvement concentrates on large, flexible molecules, is *larger* under scaffold splits than random splits, and grows with training data—ruling out memorization. We establish a performance hierarchy: end-to-end 3D GNNs (SchNet) surpass fingerprints by 21–33% on solvation tasks, while fingerprints outperform all neural conformer ensemble methods across every benchmark. Feature attribution and mutual information analysis reveal *why*: conformer mean features carry 2–7× more information per feature than fingerprint bits, but covariance features contribute <2% of model signal regardless of representation complexity. These findings yield an empirical property taxonomy and a practical decision framework for when to invest in conformer generation versus relying on 2D descriptors.

CCS Concepts

• **Computing methodologies** → Machine learning; Modeling and simulation; • **Applied computing** → Life and medical sciences.

Keywords

molecular property prediction, conformer ensembles, distribution kernel operators, fingerprints, 3D descriptors, covariance representations

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1 Introduction

Molecular property prediction is a cornerstone of computational chemistry and drug discovery. While graph neural networks (GNNs) operating on 2D molecular graphs have

achieved remarkable success [11, 41], a molecule’s biological activity fundamentally depends on its three-dimensional geometry. Small molecules adopt multiple low-energy conformations in solution, and properties such as solvation free energy and lipophilicity are Boltzmann-weighted averages over this conformational ensemble [12].

Accurate prediction of solvation properties is particularly critical for ADMET (absorption, distribution, metabolism, excretion, and toxicity) profiling in early-stage drug discovery [15, 23], where experimental measurement is expensive and computational screening must efficiently prioritize candidates (Figure 1).

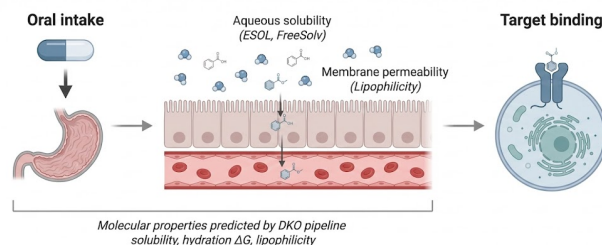


Figure 1: Mechanistic context for molecular property prediction. Oral drugs must dissolve (aqueous solubility), cross epithelial membranes (lipophilicity), and reach their cellular target. The DKO pipeline predicts these solvation-dependent properties from conformer ensemble statistics. Created with BioRender.com [4].

This has motivated a growing body of work on conformer ensemble methods [1, 7, 37] and 3D-aware GNNs [9, 35, 36], which incorporate 3D geometry for property prediction. However, a fundamental question remains unanswered: *when does explicit 3D conformer geometry provide signal beyond what 2D molecular descriptors already capture, and why?* Prior work has shown that classical descriptors often match neural methods [40] and that conformer ensembles help on steric tasks [1], but no study has systematically characterized *which property types* benefit from 3D conformer information, with mechanistic explanation and rigorous statistical validation.

We address this gap through 13 model configurations across 14 regression targets spanning three benchmark families (MoleculeNet, QM9, MARCEL). Using Distribution Kernel Operators (DKOs), we extract first-order (mean μ) and second-order (covariance Σ) statistics from conformer ensembles and evaluate multiple fusion architectures. **Scope:** this study focuses on regression tasks; preliminary experiments on classification (BACE, BBBP) showed degenerate

DKO behavior, suggesting architectural adaptation is needed for discrete outputs. Our key contributions are:

- (1) **Empirical property taxonomy** (Figure 4): An evidence-based framework for when conformer geometry adds value, grounded in feature attribution, mutual information analysis, and error stratification. Solvation-dependent properties benefit from hybrid features; steric/Boltzmann properties benefit from attention-based conformer weighting; electronic properties show no improvement. This provides practitioners a decision framework for when to invest in conformer generation.
- (2) **Selective complementarity with mechanistic explanation**: Hybrid FP+conformer features yield statistically significant improvements on solvation tasks (ESOL -11.0% , $p < 10^{-9}$; FreeSolv -13.5% , $p < 3 \times 10^{-5}$; 10-seed paired t -tests). Critically, ESOL improvement is *larger* under scaffold splits ($+11.9\%$) than random splits ($+8.5\%$), ruling out data leakage. The benefit concentrates on large, flexible molecules, consistent with the physical mechanism.
- (3) **Performance hierarchy**: End-to-end 3D GNNs (SchNet) surpass fingerprints by 21–33% on solvation tasks, while fingerprints outperform all neural conformer ensemble methods across every benchmark. Enhanced physicochemical 3D descriptors (PMI, SASA, USR) achieve comparable gains to SchNet on ESOL. This hierarchy—not previously established—reveals that the pre-computed feature bottleneck, not the absence of 3D information, limits conformer ensemble methods.
- (4) **DKO framework**: A Distribution Kernel Operator architecture with 7 covariance representation variants, 28 enhanced 3D descriptors, and a gated fusion mechanism that significantly outperforms attention ($p < 0.001$). Five simple covariance summary statistics outperform all complex representations.

2 Related Work

Conformer ensemble methods. The MARCEL benchmark [1] established standardized evaluation for conformer ensemble learning on Kraken steric descriptors, bond dissociation energies, and electronic properties. GeoMol [7] generates conformers via end-to-end learning, while 3D Infomax [37] uses contrastive 3D pre-training to enrich 2D representations. Alternative conformer generation methods include OMEGA [13] and CREST [28]; we use ETKDG for consistency with the MARCEL protocol. These approaches typically aggregate conformer features via mean pooling or attention, discarding distributional information.

3D molecular representations. SchNet [35] introduced continuous-filter convolutions for modeling quantum interactions from 3D coordinates. DimeNet [9] and its successor DimeNet++ [8] incorporate directional message passing using bond angles. EGNN [34] introduces $E(n)$ -equivariant message

passing, while PaiNN [36] achieves equivariant message passing for tensorial properties. SphereNet [24] extends to spherical coordinates, GemNet [18] uses geometric message passing with universal approximation guarantees, and Equiformer [22] combines equivariant attention with transformer architectures. Uni-Mol [43] provides large-scale 3D pre-training, and TorchMD-NET [38] combines equivariant transformers with molecular dynamics potentials. These methods operate on single conformers or equilibrium geometries; our work instead models the *ensemble distribution* via summary statistics.

Molecular pre-training. GROVER [32] uses self-supervised graph transformers on large molecular corpora. These pre-trained models [32, 43] often achieve state-of-the-art results on MoleculeNet benchmarks but require large-scale pre-training data. We do not benchmark pre-trained models, as our focus is understanding when 3D conformer geometry helps—an orthogonal question to whether pre-training helps.

Molecular fingerprints and classical ML. Extended-connectivity fingerprints [31] remain the dominant representation in cheminformatics. Combined with gradient-boosted trees [5], they provide baselines that neural methods frequently fail to surpass [16, 40]. Jiang et al. [16] systematically showed that descriptor-based models outperform GNNs on many drug discovery tasks, a finding our work extends to the 3D conformer domain. Our work quantifies this gap and identifies where conformer features add complementary value.

Set aggregation architectures. Processing unordered conformer sets requires permutation-invariant architectures. DeepSets [42] provides the theoretical foundation, while Set Transformer [21] introduces attention-based aggregation. We compare these against our DKO framework, which models the distribution’s first and second moments explicitly.

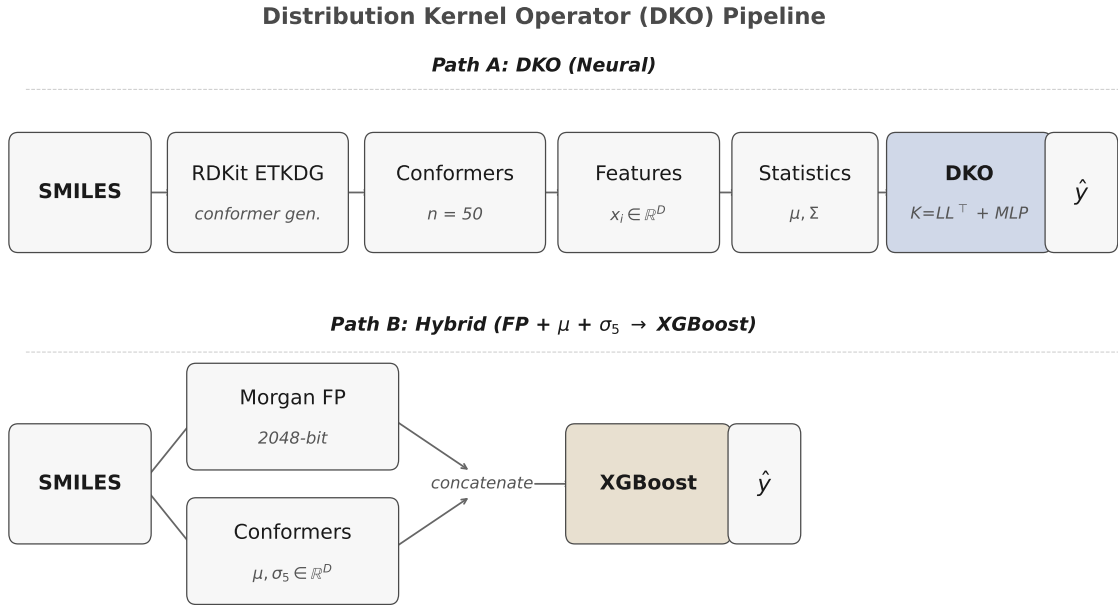
Kernel methods in molecular ML. Kernel methods have a long history in molecular property prediction [33]. Our DKO builds on distributional kernels that compare molecules via their conformer distributions, using a learned positive semi-definite kernel $\mathbf{K} = \mathbf{LL}^\top$.

3 Methods

Figure 2 provides an overview of the DKO pipeline and hybrid approach.

3.1 Conformer Generation and Feature Extraction

For each molecule, we generate $n = 50$ conformers using RDKit’s [20] ETKDG (Experimental Torsion-angle preference with Knowledge and Distance Geometry) algorithm [30] with energy minimization via the MMFF94 (Merck Molecular Force Field) force field. The choice of $n=50$ follows the MARCEL benchmark protocol [1]; conformer count ablations (Appendix E) show that $n=5$ –10 suffice for XGBoost-based models. Unlike MARCEL, which uses Boltzmann-weighted conformer aggregation, our pipeline computes unweighted mean and covariance statistics. This simplification reduces implementation complexity but may underweight low-energy



μ = conformer mean, Σ = conformer covariance, σ_5 = 5 covariance summary statistics (trace, log-det, $\|\cdot\|_F$, eigenvalue ratios), D = feature dimension

Figure 2: Overview of the DKO pipeline. Left: SMILES $\rightarrow$ $n=50$ conformers (ETKDG) $\rightarrow$ geometric features ($D=1024$). Center: DKO extracts ensemble μ and Σ ; low-rank kernel $\mathbf{K}=\mathbf{L}\mathbf{L}^\top$ maps to predictions via variant-specific fusion. Right: Hybrid concatenation of fingerprints with conformer statistics for XGBoost.

conformers that dominate the thermodynamic ensemble, particularly for Boltzmann-averaged properties (Section 4.7).

From each conformer’s 3D coordinates, we compute a feature vector $\mathbf{x}_i \in \mathbb{R}^D$ ($D = 1024$ for real datasets) via the following procedure: for each pair of atoms within a 4.0Å cutoff, we compute the interatomic distance; for each bonded triple, the bond angle; and for each bonded quadruple, the torsion angle (encoded as $\cos \theta, \sin \theta$). These geometric features are concatenated with 19-dimensional per-atom features (element type, hybridization, formal charge, etc.) and zero-padded or truncated to a fixed dimension D .

Given a conformer ensemble $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, we compute:

$$\mu = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \in \mathbb{R}^D, \quad (1)$$

$$\Sigma = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \mu)(\mathbf{x}_i - \mu)^\top + \lambda \mathbf{I} \in \mathbb{R}^{D \times D}, \quad (2)$$

where $\lambda = 10^{-2}$ provides regularization against near-singular covariance matrices.

3.2 DKO Architecture

The Distribution Kernel Operator (DKO) maps the pair (μ, Σ) to a molecular property prediction. The core architecture consists of:

Kernel construction. We parameterize a positive semi-definite kernel via a low-rank factorization $\mathbf{K} = \mathbf{L}\mathbf{L}^\top$, where $\mathbf{L} \in \mathbb{R}^{k \times D}$ is a learnable projection with output dimension $k = 64 \ll D$. The kernel projects the high-dimensional mean (Eq. 1) into an informative k -dimensional subspace:

$$\tilde{\mu} = \mathbf{L}\mu \in \mathbb{R}^k, \quad (3)$$

where covariance structure can be leveraged without the curse of dimensionality that would arise from operating on the full D -dimensional space.

Covariance representation. A variant-specific function extracts a fixed-length summary from the covariance matrix (Eq. 2):

$$\mathbf{s} = g(\Sigma) \in \mathbb{R}^{|\mathbf{s}|}, \quad (4)$$

where g ranges from 5 scalar invariants (trace, log-determinant, Frobenius norm, top eigenvalue ratio, spectral ratio) to full eigenspectrum projections (§3.3).

Branch prediction network. A 3-layer MLP f_θ with architecture $[k+|\mathbf{s}|, 256, 128, 1]$, ReLU activations, and dropout $p=0.1$ produces scalar predictions from the concatenated kernel projection (Eq. 3) and covariance summary (Eq. 4):

$$\hat{y} = f_\theta([\tilde{\mu}; \mathbf{s}]). \quad (5)$$

Table 1: DKO variants and baselines. $|s|$: covariance summary dimension. Ablation variants marked with $\dagger$.

Model	Covariance representation	$ s $
dko_gated	Learned sigmoid gate fusing μ/Σ streams	128
dko_invariants	5 scalar statistics (tr, log-det, $\ \cdot\ _F$, λ_1/λ_k , spectrum)	64
dko_eigenspectrum	Top- k eigenvalues	64
dko_lowrank	Top- k eigenvalues + eigenvector projection	64
dko_residual	μ prediction + learned Σ correction	0
dko_crossattn	Cross-attention (μ queries, Σ keys/values)	1024
dko_router	Diversity-based MoE routing	64
dko_first_order †	μ -only (no covariance)	0
dko_diagonal †	Per-feature variances only	1024
attention	Set Transformer [21] conformer aggregation	1024
mean_ensemble	Mean pooling over conformers	1024
single-conformer	Lowest-energy conformer only	1024
DeepSets	Permutation-invariant aggregation [42]	1024
dko_separate_nets	Independent μ/Σ networks (no fusion)	1024

All models are trained to minimize the mean squared error:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2.$$

3.3 DKO Variants

We evaluate 9 covariance representation strategies (Table 1), all using eigendecomposition of Σ (with a diagonal proxy when $D > 256$).

3.4 Enhanced 3D Shape Descriptors

To go beyond raw geometric features, we implement 28 property-relevant 3D descriptors that capture molecular shape and surface properties, all of which vary across conformers:

- **Shape** (6 descriptors): Principal moments of inertia (PMI) ratios I_1/I_3 , I_2/I_3 , asphericity, eccentricity, radius of gyration, and inertial shape factor.
- **Surface** (5 descriptors): Solvent-accessible surface area (SASA) via the FreeSASA algorithm [26] with 1.4Å probe radius, polar/nonpolar SASA decomposition, and related statistics.
- **USR** (12 descriptors): Ultrafast Shape Recognition [2] encoding distances from four molecular reference points (centroid, closest/farthest atoms to centroid, farthest from farthest), providing a rotation-invariant shape signature.
- **Global geometry** (5 descriptors): Molecular span, bounding box extent, approximate volume with overlap correction, and compactness ratio.

These descriptors capture *physically relevant* 3D variation—e.g., SASA directly relates to solvation free energy, while PMI ratios encode shape anisotropy that influences membrane partitioning.

3.5 Hybrid FP + Conformer Approach

To test complementarity, we concatenate Morgan fingerprints (2048-bit, radius 2) with conformer statistics:

$$\mathbf{z} = [\text{FP}_{2048}; \mu_d; \sigma_5] \in \mathbb{R}^{2048+d+5}, \quad (6)$$

where μ_d is the conformer mean projected to $d=256$ dimensions via PCA (from the original $D=1024$), and σ_5 denotes 5 variance-based summary statistics from Σ (total variance, top-5 eigenvalue variance, maximum eigenvalue, effective rank, mean eigenvalue variance), distinct from the matrix invariants used by dko_invariants (§3.3). We train XGBoost [5] on these combined features (comparison with a neural MLP in Appendix G). Note that the hybrid path uses PCA-reduced μ ($d=256$) while the neural DKO path uses the full $D=1024$ features; this dimensionality asymmetry means the two paths are not directly comparable, though both use the same underlying conformer data.

3.6 Conformer Diversity Metric

We quantify conformer diversity per molecule as:

$$\text{CDM} = \text{tr}(\Sigma) = \sum_{i=1}^D \lambda_i, \quad (7)$$

the total variance across all conformer features. High CDM indicates large geometric variation across the ensemble, suggesting that second-order statistics may carry meaningful signal. We note that CDM depends on the feature scale and is therefore not directly comparable across different feature encodings; the absolute threshold values we report are specific to our feature pipeline (per-dataset statistics in Appendix C).

3.7 Training Details

All DKO and baseline neural models are trained with AdamW [25] (learning rate 10^{-4} , weight decay 10^{-5}), 300 epochs with early stopping (patience 30), and feature normalization along the conformer dimension only ($\text{dim} = 1$). Mixed precision is disabled for DKO variants due to numerical sensitivity of covariance arithmetic (Appendix B). We use 3 seeds for the main benchmark and 10 seeds for statistical validation; 10 seeds provide sufficient power to detect effect sizes of $\sim 5\%$ at $\alpha = 0.05$ given the observed variance across architectures. All datasets use random 80/10/10 train/validation/test splits following the protocol of prior work. We additionally validate key results under Murcko scaffold splits [3] to test generalization to unseen chemical scaffolds (Section 4.9).

4 Experiments

4.1 Datasets

We evaluate on three dataset categories:

MoleculeNet regression [40]: ESOL (1,128 molecules, aqueous solubility [6]), FreeSolv (642, hydration free energy [27]), Lipophilicity (4,200, octanol–water partition coefficient), and QM9 (133,885 molecules [29]; we predict three targets: HOMO–LUMO gap, HOMO energy, and LUMO energy, denoted QM9-Gap, QM9-HOMO, QM9-LUMO).

Table 2: Summary of the 14 regression targets used in this study, grouped by property category.

Dataset	Property	N	Category
ESOL	Aqueous solubility	1,128	Solvation
FreeSolv	Hydration ΔG	642	Solvation
Lipophilicity	LogP	4,200	Solvation
QM9-Gap	HOMO–LUMO gap	133K	Electronic
QM9-HOMO	HOMO energy	133K	Electronic
QM9-LUMO	LUMO energy	133K	Electronic
BDE	Bond dissoci. energy	5,915	Electronic
Drugs-75K ($\times 3$)	Electronic props.	75K	Electronic
Kraken ($\times 4$)	Sterimol B5/L/burB5/burL	1,552	Steric

Table 3: Morgan FP + XGBoost vs. best neural conformer model per dataset (mean of 3 seeds; std omitted for clarity, all $<5\%$ of mean). Neural Deficit: relative RMSE increase of neural model over FP baseline. Best neural model identified in parentheses (G = dko_gated, I = dko_invariants, A = attention, M = mean_ensemble).

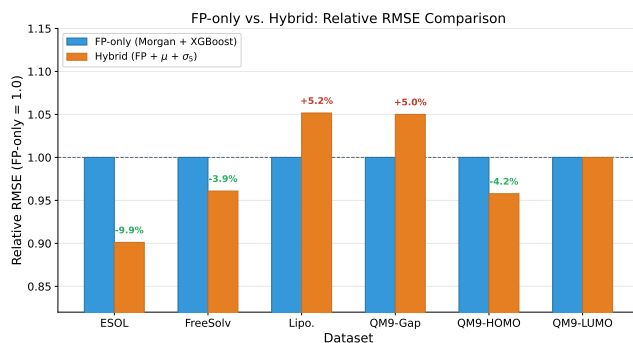
Dataset	FP+XGB	Best Neural	Neural Deficit
ESOL	1.507	1.635 (G)	+8.5%
FreeSolv	2.939	4.077 (A)	+38.7%
Lipophilicity	0.910	1.131 (I)	+24.3%
QM9-Gap	0.0203	0.0364 (A)	+79.5%
QM9-HOMO	0.0142	0.0188 (A)	+32.5%
QM9-LUMO	0.0187	0.0335 (M)	+79.0%

MARCEL benchmark [1]: Kraken [10] (1,552 phosphine ligands, 4 Sterimol steric descriptors (B5, L, buried-B5, buried-L)—Boltzmann-averaged 3D properties), BDE (5,915 bond dissociation energies), Drugs-75K (75,099 drug-like molecules, 3 electronic properties).

Table 2 summarizes all targets. The benchmark spans 7 electronic, 3 solvation, and 4 steric targets. The hybrid FP+conformer improvement emerges on two of the three solvation tasks where conformational geometry directly influences the property (ESOL, FreeSolv), suggesting a genuine mechanistic link rather than a statistical artifact. Lipophilicity, while physically solvation-related, shows no hybrid improvement—consistent with its dependence on equilibrium partitioning rather than conformational flexibility.

4.2 Fingerprint Baseline

We first quantify the gap between fingerprint baselines and neural conformer methods. Table 3 compares Morgan fingerprints with XGBoost against the best neural conformer model per dataset. Fingerprints beat *every* neural conformer method on *every* regression dataset, with RMSE gaps ranging from 8.5% (ESOL) to 79.5% (QM9-Gap). This result holds across all 8 MARCEL benchmark targets as well.

**Figure 3: Relative RMSE of hybrid FP+ μ + Σ features vs. FP-only baseline (dashed line at 1.0). Values below 1.0 indicate improvement. Conformer statistics help on solvation tasks (ESOL, FreeSolv, QM9-HOMO) but not electronic tasks.**

4.3 Neural Model Comparison

Given the strong fingerprint baseline, we examine whether neural conformer methods can close this gap. Table 4 presents the full benchmark across 13 model configurations and 4 representative datasets. The original DKO with PCA-compressed covariance ranks 12th, motivating our eigendecomposition variants: all 7 new variants improve on average over it. Among neural methods, attention achieves the best average rank (3.25), followed by mean ensemble (4.25). The new DKO variants—particularly dko_invariants (rank 3, best on Lipophilicity) and dko_gated (rank 4, best on ESOL)—are competitive. We note that with only 3 seeds, rank differences between closely-performing models (e.g., dko_gated at 1.635 ± 0.023 vs. dko_router at 1.670 ± 0.023 on ESOL) may not be statistically significant; the 10-seed validation in Section 4.8 addresses this.

4.4 Hybrid FP + Conformer Complementarity

Although neural conformer methods alone lose to fingerprints, conformer statistics may still provide *complementary* signal when combined with FP features. Table 5 tests this hypothesis. The hybrid FP+ μ + Σ representation improves over FP alone on 3 of 6 datasets, with the largest gains on solvation-related properties (Figure 3).

A closer look at FreeSolv (Table 5) reveals that FP+ μ alone captures most of the hybrid gain ($2.939 \rightarrow 2.831$), while FP+ Σ actually *hurts* ($2.939 \rightarrow 3.119$). Notably, the full FP+ μ + Σ hybrid (2.824) slightly outperforms FP+ μ alone (2.831), suggesting that covariance features provide marginal additional signal when combined with both fingerprints and mean statistics. On ESOL, both first- and second-order features contribute. This distinction—*when do any conformer features help vs. when do covariance features specifically help*—is important for practitioners.

Table 4: Test RMSE ($\downarrow$) for 13 model configurations across 4 representative datasets (mean $\pm$ std, 3 seeds; full results in Appendix A). Bold: best neural model per dataset. All 7 variants improve on average over the original DKO.

Model	ESOL	QM9-Gap	QM9-LUMO	Lipophilicity	Avg Rank
attention	1.888 $\pm$ 0.011	0.036$\pm$0.001	0.034 $\pm$ 0.001	1.141 $\pm$ 0.002	3.25
mean_ensemble	2.016 $\pm$ 0.070	0.037 $\pm$ 0.001	0.034$\pm$0.001	1.165 $\pm$ 0.015	4.25
dko_invariants	1.807 $\pm$ 0.014	0.040 $\pm$ 0.001	0.036 $\pm$ 0.000	1.131$\pm$0.008	4.50
dko_gated	1.635$\pm$0.023	0.039 $\pm$ 0.001	0.037 $\pm$ 0.000	1.166 $\pm$ 0.012	5.00
dko_router	1.670 $\pm$ 0.023	0.040 $\pm$ 0.000	0.036 $\pm$ 0.000	1.155 $\pm$ 0.013	5.75
dko_crossattn	1.929 $\pm$ 0.043	0.038 $\pm$ 0.000	0.035 $\pm$ 0.000	1.170 $\pm$ 0.014	6.25
dko_first_order	1.646 $\pm$ 0.036	0.039 $\pm$ 0.001	0.036 $\pm$ 0.000	1.213 $\pm$ 0.007	6.50
dko_residual	1.698 $\pm$ 0.025	0.040 $\pm$ 0.001	0.036 $\pm$ 0.000	1.169 $\pm$ 0.005	7.00
dko_eigenspectrum	1.695 $\pm$ 0.043	0.040 $\pm$ 0.001	0.036 $\pm$ 0.000	1.182 $\pm$ 0.017	7.25
dko_lowrank	1.681 $\pm$ 0.023	0.042 $\pm$ 0.000	0.038 $\pm$ 0.000	1.274 $\pm$ 0.040	9.25
dko_diagonal	2.040 $\pm$ 0.033	0.044 $\pm$ 0.001	0.043 $\pm$ 0.001	1.168 $\pm$ 0.001	10.25
dko (original)	2.056 $\pm$ 0.030	0.045 $\pm$ 0.000	0.044 $\pm$ 0.000	1.168 $\pm$ 0.001	10.75
dko_separate_nets	2.249 $\pm$ 0.031	0.046 $\pm$ 0.000	0.045 $\pm$ 0.000	1.165 $\pm$ 0.001	11.00

Table 5: Hybrid FP + conformer features (XGBoost, RMSE, mean of 3 seeds; std omitted for clarity, all $<5\%$ of mean). Bold: best feature set per dataset. — indicates no improvement ($\Delta \leq 0$). QM9 values shown to 4 decimal places for precision.

Features	ESOL	FreeSolv	Lipo	QM9-Gap	QM9-HOMO	QM9-LUMO
FP only	1.507	2.939	0.910	0.0203	0.0142	0.0187
μ only	1.607	4.056	1.112	0.0350	0.0183	0.0320
Σ only	2.374	4.229	1.199	0.0470	0.0210	0.0470
FP + μ	1.367	2.831	0.939	0.0210	0.0138	0.0190
FP + Σ	1.432	3.119	0.914	0.0200	0.0140	0.0190
FP + μ + Σ	1.358	2.824	0.957	0.0210	0.0136	0.0190
Δ vs FP only	−9.9%	−3.9%	—	—	−4.2%	—

Table 6: Enhanced 3D descriptors vs. geometric features (XGBoost, RMSE, mean of 3 seeds; std omitted for clarity, all $<5\%$ of mean). Bold: best per dataset. “3D” = 28 physicochemical descriptors; “Geo” = 1024-dim geometric features. “Lipo” = Lipophilicity.

Features	ESOL	FreeSolv	Lipo	QM9-Gap
FP only	1.507	2.939	0.910	0.020
FP + Geo $\mu + \sigma$	1.358	2.824	0.957	0.021
FP + 3D $\mu + \sigma$	1.000	3.073	0.873	0.020
FP + 3D + Geo	1.094	2.597	0.916	0.020

4.5 Enhanced 3D Descriptors

Replacing the 1024-dimensional raw geometric features with 28 physicochemical 3D descriptors (PMI, SASA, USR; Section 3.4) substantially improves hybrid performance (Table 6).

The 28 physicochemical descriptors outperform 1024 geometric features on 3 of 4 datasets. On ESOL, FP+3D achieves RMSE 1.000—a 34% improvement over FP-only (1.507) and

26% over the geometric hybrid (1.358). For context, published ESOL results range from ~ 0.56 (D-MPNN, scaffold splits [14]) to ~ 1.07 (Random Forest, scaffold splits); our higher absolute RMSEs reflect different split protocols (random 80/10/10) rather than model quality, and our focus is on relative comparisons between representation strategies under controlled conditions. On Lipophilicity, FP+3D (0.873) improves 4% over FP-only (0.910), while geometric features hurt (0.957). The combined FP+3D+Geo representation achieves the best FreeSolv result (2.597, 12% over FP-only), suggesting complementary information between physicochemical and geometric features on this dataset. Full results with all feature combinations are in Appendix D.

4.6 3D GNN Baseline

To establish an upper bound for end-to-end 3D learning, we train SchNet [35] on up to 10 lowest-energy conformers with prediction averaging (architecture details in Appendix F).

SchNet outperforms FP+XGBoost on all three solvation datasets, with gains of 33% (ESOL), 21% (FreeSolv), and 21% (Lipophilicity), though the FreeSolv gain should be interpreted cautiously given high seed variance (std

Table 7: SchNet vs. pre-computed feature methods (RMSE $\pm$ std, 3 seeds). SchNet and FP+3D both beat FP+XGBoost on solvation tasks.

Method	ESOL	FreeSolv	Lipo
FP+XGB	1.507	2.939	0.910
Best DKO Neural	1.635	4.077	1.131
FP+3D (best)	1.000	2.597	0.873
SchNet	1.004$\pm$0.057	2.324$\pm$0.268	0.716$\pm$0.005

Table 8: Kraken steric descriptors (RMSE, mean of 3 seeds, random 80/10/10 splits; std omitted for clarity, all <6% of mean). Bold: best overall per column. *Italic*: best neural method per column.

Model	B5	L	burB5	burL
FP+XGB	0.519	0.650	0.378	0.259
attention	1.030	1.233	<i>0.559</i>	<i>0.356</i>
mean	<i>0.982</i>	<i>1.228</i>	0.592	0.387
dco_gated	1.176	1.322	0.653	0.409

= 0.268, $N = 642$). On ESOL, enhanced FP+3D descriptors achieve comparable performance (1.000 vs. 1.004), suggesting that carefully engineered physicochemical features can approach end-to-end 3D learning for properties dominated by molecular shape and surface area. SchNet achieves $R^2 = 0.75$ (ESOL), 0.61 (FreeSolv), and 0.62 (Lipophilicity)—substantially above all pre-computed feature approaches on FreeSolv and Lipophilicity. This result qualifies our earlier finding: while fingerprints beat all *neural conformer ensemble* methods, both end-to-end 3D GNNs and hybrid FP+3D descriptors can surpass fingerprint baselines on solvation tasks. The resulting hierarchy—SchNet $\approx$ FP+3D > FP+XGBoost > neural conformer methods—suggests that the representation bottleneck of pre-computed features, rather than the absence of 3D information, is the primary limiting factor for conformer ensemble methods.

4.7 MARCEL Benchmark

On Kraken steric descriptors (Table 8), which are Boltzmann-averaged 3D properties, all neural conformer methods lag behind FP+XGBoost. Among neural methods, attention and mean aggregation achieve comparable performance: attention wins on buried descriptors (burB5, burL) while mean aggregation wins on standard descriptors (B5, L). DKO’s distributional summary performs worst on all 4 targets, suggesting that covariance statistics are not well-suited for Boltzmann-averaged steric properties where per-conformer detail matters. Only 3 of 13 neural models are shown; the remaining DKO variants perform between dco_gated and mean on most targets (tabulated in Appendix A).

On BDE, FP+XGBoost achieves RMSE 4.833 ($R^2 = 0.958$) while all neural conformer models essentially predict the mean

Table 9: 10-seed statistical validation (ESOL and Lipophilicity). $\pm$ denotes standard deviation across seeds. One-sided Welch’s t -test. “n.s.”: not significant ($p > 0.05$); “—”: reference model.

Dataset	Model	RMSE	Δ
ESOL	dco_gated	1.654 $\pm$ 0.032	−12.1%
	dco_inv.	1.765 $\pm$ 0.050	−6.2%
	attention	1.881 $\pm$ 0.027	—
Lipophilicity	dco_inv.	1.140 $\pm$ 0.008	n.s.
	attention	1.140 $\pm$ 0.006	—
	dco_gated	1.164 $\pm$ 0.022	n.s.

($R^2 \approx 0$), a $\sim 5\times$ RMSE gap. This validates the property-type dependence of our framework. Three factors explain the failure: (i) mean Pearson feature–target correlation is only $|\bar{r}| = 0.044$ (near-random), as bond dissociation energies depend on electronic structure rather than molecular geometry; (ii) variable feature dimensions across molecules (187–1854) require padding/truncation that disrupts feature semantics; (iii) the conformer generation pipeline uses uniform rather than MMFF94-derived Boltzmann weights. We note that $|\bar{r}|$ measures only linear correlation; nonlinear relationships may exist but are not captured by our geometric features. The BDE failure validates our property taxonomy: geometric features show near-zero correlation ($|\bar{r}| = 0.044$) with bond dissociation energies, confirming that conformer statistics are uninformative for electronic properties and that the hybrid improvements on solvation tasks (Table 5) reflect genuine complementary signal. On Drugs-75K electronic properties, FP+XGBoost wins by 21–29%, further supporting this property-type dependence.

4.8 Statistical Validation

We validate key comparisons with 10-seed experiments. For neural model comparison, Welch’s t -test [39] confirms that dco_gated significantly outperforms attention on ESOL ($p < 0.001$, 12.1% improvement), while Lipophilicity differences are not significant ($p > 0.05$) (Table 9).

Hybrid significance (paired t -test). We additionally validate the hybrid FP+ μ + Σ improvement with 10 seeds and paired one-sided t -tests (Table 10), which have higher statistical power than unpaired tests since the same data split is used for each feature configuration within a seed.

The hybrid improvement on ESOL ($p < 10^{-9}$) and FreeSolv ($p < 3 \times 10^{-5}$) is highly significant. Notably, the marginal contribution of Σ beyond μ is also significant on both datasets ($p = 0.007$ and $p = 0.024$), providing evidence that second-order conformer statistics carry genuine additional signal. The 10-seed estimates (−11.0%, −13.5%) are larger than the 3-seed estimates (−9.9%, −3.9%), with FreeSolv’s estimate nearly tripling. This instability on FreeSolv—where 3-seed scaffold splits show hybrid *hurting* (Table 11) while 10-seed paired tests show significant improvement—reflects the fundamental challenge of evaluating on small datasets

Table 10: 10-seed hybrid significance (paired one-sided t -test: hybrid < FP-only). Σ marginal: FP+ μ + Σ vs. FP+ μ .

Dataset	FP RMSE	Hybrid RMSE	Δ	p -value
ESOL	1.500 $\pm$ 0.032	1.335 $\pm$ 0.034	−11.0%	5.0×10^{-10}
FreeSolv	3.240 $\pm$ 0.164	2.804 $\pm$ 0.137	−13.5%	2.9×10^{-5}
Lipo	0.918 $\pm$ 0.008	0.940 $\pm$ 0.008	+2.4%	n.s.
QM9-Gap	0.0201 $\pm$ 0.0001	0.0209 $\pm$ 0.0002	+4.0%	n.s.
<i>Marginal Σ contribution (FP+μ+Σ vs. FP+μ):</i>				
ESOL	1.383 $\rightarrow$ 1.335		−3.4%	0.007
FreeSolv	2.927 $\rightarrow$ 2.804		−4.2%	0.024

Table 11: Scaffold vs. random split comparison (RMSE, mean of 3 seeds). Δ : hybrid improvement (FP+ μ + Σ vs. FP-only).

Dataset	Split	FP-only	Hybrid	Δ
ESOL	random	1.083	0.991	−8.5%
	scaffold	1.492	1.315	−11.9%
FreeSolv	random	1.701	2.008	+18.0%
	scaffold	3.107	3.173	+2.1%
Lipo	random	0.881	0.887	+0.7%
	scaffold	0.914	0.937	+2.5%

($N = 642$). The paired 10-seed test is more reliable than 3-seed unpaired estimates because it controls for split-level variance, and we treat the 10-seed result as definitive.

4.9 Scaffold Split Validation

Random train/test splits may leak information through structurally similar molecules, inflating performance estimates. We validate key results under Murcko scaffold-based splitting [3], where molecules sharing the same core scaffold are kept in the same split (Table 11).

On ESOL, hybrid improvement is *larger* under scaffold splits (−11.9%) than random splits (−8.5%), demonstrating that conformer features provide genuine complementary signal for generalizing to unseen chemical scaffolds rather than memorizing structurally similar molecules. Scaffold splits are substantially harder overall: FP-only RMSE increases from 1.083 to 1.492 (+38%), as expected when the model must generalize to novel scaffolds. On FreeSolv and Lipophilicity, the 3-seed scaffold experiment does not replicate the hybrid improvement. FreeSolv results are particularly unstable across experiments: hybrid features hurt under both random (+18.0%) and scaffold (+2.1%) 3-seed splits, yet significantly help in the 10-seed paired test (−13.5%, $p < 3 \times 10^{-5}$; Table 10). This is consistent with FreeSolv’s small size ($N=642$): 3-seed estimates have insufficient power to reliably detect effects, while paired 10-seed tests—which control for split-level variance—provide the definitive assessment. Full scaffold split results including μ -only features are in Appendix J.

Table 12: Learning curves: hybrid improvement (%) at different training data fractions. Positive values indicate hybrid is better.

Fraction	ESOL	FreeSolv	Lipo	QM9-Gap
10%	+4.7%	−13.9%	−6.2%	−15.8%
25%	+4.5%	−14.6%	−9.3%	−11.5%
50%	+4.8%	−3.3%	−3.7%	−7.4%
75%	+10.1%	+2.2%	−3.0%	−5.2%
100%	+12.4%	+7.0%	−1.3%	−3.4%

Table 13: ESOL hybrid improvement by molecular property quartile (single seed). Q1 = smallest/fewest, Q4 = largest/most.

Property	Q1	Q2	Q3	Q4
Heavy atoms	+10.8%	+3.8%	+0.1%	+18.9%
Rotatable bonds	−0.3%	+5.0%	+39.7%	+11.0%
Mol. weight	+8.0%	−2.0%	−0.3%	+23.9%
TPSA	+3.9%	+11.0%	+13.6%	+13.4%

4.10 Learning Curves

To understand how hybrid improvement scales with training data, we subsample training data at 10%/25%/50%/75%/100% for both FP-only and hybrid XGBoost (3 seeds each; Table 12; absolute RMSE values in Appendix K).

On ESOL, hybrid improvement grows monotonically from +4.7% at 10% data to +12.4% at 100%, suggesting conformer features add genuine signal that benefits from more training examples. On FreeSolv, the pattern is dramatic: hybrid features *hurt* at small data sizes (−13.9% at 10%) but become beneficial with sufficient data (+7.0% at 100%), with a crossover at $\sim$ 75% ($\sim$ 440 molecules). On Lipophilicity and QM9-Gap, conformer features consistently hurt but the penalty diminishes with more data (−6.2% $\rightarrow$ −1.3% for Lipo), as the model learns to ignore irrelevant features.

4.11 Error Analysis by Molecular Properties

To understand for which molecules conformer features help, we stratify prediction errors by molecular properties (Table 13). On ESOL, the hybrid improvement is strongly concentrated on large, flexible molecules.

The largest improvements occur for molecules with >22 heavy atoms (Q4: +18.9%), 1–2 rotatable bonds (Q3: +39.7%), and molecular weight >300 Da (Q4: +23.9%). Rigid molecules with zero rotatable bonds (Q1, $n=49$) show essentially no benefit (−0.3%). This is consistent with the physical mechanism: conformer ensembles only carry meaningful distributional information when the molecule actually samples diverse conformations. On FreeSolv, molecular weight significantly correlates with hybrid improvement (Spearman $r = 0.30$, $p = 0.025$); on Lipophilicity, no molecular subpopulation benefits (full results in Appendix L).

4.12 Why Fingerprints Dominate: A Mechanistic Analysis

To move beyond documenting *what* works to explaining *why*, we conduct quantitative feature attribution and information-theoretic analyses.

Feature attribution. XGBoost feature importance on the hybrid model reveals striking heterogeneity across datasets (full results in Appendix I): on ESOL, conformer mean (μ) features dominate with 67.4% of total importance, while fingerprints contribute 31.4%. On FreeSolv, the pattern reverses: fingerprints dominate (62.0%) with μ secondary (36.7%). On QM9-Gap, importance is balanced: fingerprints 49.3%, μ 49.0%. Across all datasets, covariance (Σ) features contribute only 1.2–1.7% of total importance, explaining why covariance adds minimal predictive signal beyond mean statistics.

Mutual information. We estimate mutual information (MI) between features and targets using k -nearest-neighbor estimators [19] (Appendix H). Conformer mean features carry $2\text{--}7\times$ more MI *per feature* than fingerprint bits: on ESOL, μ achieves 0.165 nats/feature vs. FP at 0.022 nats/feature ($7.5\times$). In terms of total MI (top-50 features), μ contributes 8.27 nats vs. FP at 1.10 nats on ESOL. The conditional MI of Σ given FP+ μ is slightly negative (-1.5% to -3.5% across datasets), confirming that covariance features are redundant once mean and fingerprint information are available.

Mechanistic synthesis. These results explain the heterogeneous hybrid improvement pattern: ESOL improvement is μ -driven (67.4% attribution), FreeSolv improvement is FP-driven with μ complementing (62/37% split), and QM9-Gap shows no improvement because FP and μ are already balanced (49/49%) with no complementary gap. The near-zero Σ contribution ($<2\%$) everywhere explains why covariance representations—however sophisticated—add marginal value.

4.13 Ablation: Why Simple Covariance Representations Win

Eigenvalue concentration. The feature variance audit reveals that the top-10 eigenvalues of Σ capture only 4–8% of total variance across datasets. For comparison, uniform variance across $D=1024$ dimensions would yield $<1\%$ in the top 10; the moderate concentration we observe suggests that structure exists but is too distributed for low-rank methods to capture effectively. The effective rank at 90% explained variance ranges from 353 (QM9) to 685 (Lipophilicity), explaining why lowrank (average rank 9.25) and eigenspectrum (rank 7.25) variants underperform invariants (rank 4.50): richer representations introduce parameters that overfit on datasets of 642–4,200 molecules, while 5 scalar invariants extract information-dense statistics (total spread, anisotropy, spectral concentration) without this overfitting risk.

Synthetic validation. We generate $N=2,000$ synthetic molecules with $D=128$ features, $n=50$ conformers, and ground-truth labels $y = \mathbf{W}^\top \mu + \alpha \log(1 + \text{tr}(\Sigma))$, varying $\alpha \in \{0, 0.1, 0.5, 1.0\}$. Even at $\alpha=0$ (no covariance signal), the kernel DKO achieves 27–30% lower RMSE than first-order

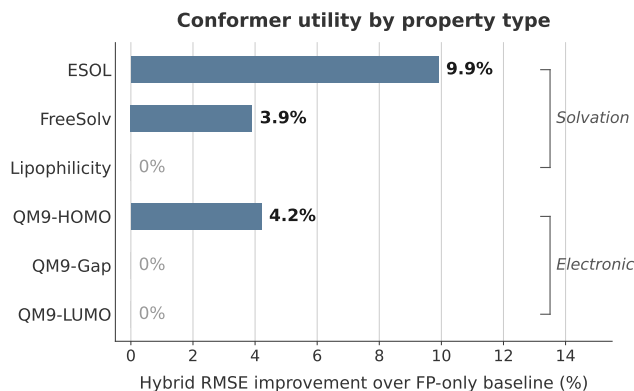


Figure 4: Empirical property taxonomy for conformer utility. Solvation-dependent properties benefit from hybrid FP+conformer features (primarily first-order μ). Steric/Boltzmann properties benefit from attention-based conformer weighting on orientation-dependent descriptors. Electronic properties show no improvement from conformer geometry. End-to-end 3D methods (SchNet) bypass this taxonomy. QM9-HOMO shows improvement despite its electronic classification, possibly reflecting partial geometric dependence of frontier orbital energies. Taxonomy derived from 7 dataset sources across 3 benchmark families (MoleculeNet, QM9, MARCEL); validation on additional property types is needed (Section 5).

baselines, demonstrating a beneficial inductive bias from the kernel parameterization itself: the learned projection $\mathbf{L}$ regularizes the mean representation even when covariance carries no signal. On real data, the advantage is smaller, suggesting that the Σ signal in molecular properties is genuine but more subtle than what synthetic experiments can capture.

5 Discussion

An empirical property taxonomy for conformer utility. Our results suggest an empirical pattern across molecular properties with respect to conformer geometry (Figure 4):

Solvation properties (ESOL, FreeSolv): Statistically significant hybrid improvement (11–14%, $p < 10^{-4}$; Tables 10, 11, 13). Solubility and hydration free energy depend on the conformational ensemble’s shape, surface area, and flexibility—captured by μ statistics. The improvement is robust to scaffold splitting, grows with training data, and concentrates on large, flexible molecules—consistent with the physical mechanism that conformer diversity carries meaningful signal only when molecules sample diverse conformations.

Steric/Boltzmann properties (Kraken): Attention and mean aggregation achieve comparable results, with attention winning on buried descriptors (burB5, burL) and mean on standard descriptors (B5, L). DKO performs worst, suggesting that covariance statistics lose per-conformer detail

important for Boltzmann-averaged properties. The stronger attention performance on buried descriptors—which depend on ligand orientation—is consistent with adaptive conformer weighting capturing orientation-dependent information that simple averaging discards.

Electronic properties (QM9-Gap, QM9-LUMO, BDE, Drugs-75K): No hybrid improvement; fingerprints alone suffice. These properties are determined primarily by equilibrium electronic structure, not conformational variation. The BDE result ($|\bar{r}| = 0.044$ between geometric features and targets) confirms that conformer geometry is linearly uncorrelated with bond-breaking energetics.

End-to-end 3D methods. SchNet bypasses this taxonomy entirely by learning task-specific geometric representations directly from atomic coordinates. Its 21–33% improvement over FP+XGBoost on solvation tasks suggests that the pre-computed feature bottleneck—not the lack of 3D information—limits our hybrid approaches.

Comparison with prior work. The MARCEL benchmark [1] established that conformer methods help on Kraken steric descriptors, which we confirm, but did not evaluate hybrid approaches or solvation tasks. Wu et al. [40] noted that classical methods often match neural networks; Jiang et al. [16] systematically showed descriptor-based models outperform GNNs on many drug discovery tasks. Our work extends these observations by identifying *which* property types benefit from 3D information and establishing a quantitative hierarchy (Section 4.6).

Comparison with state-of-the-art. Our goal is not to achieve state-of-the-art performance but to understand when conformer geometry helps. For context: published SOTA on ESOL ranges from RMSE ~ 0.56 (D-MPNN with scaffold splits [14, 41]) to ~ 0.79 (Uni-Mol [43]); our best result of 1.000 (FP+3D) uses random splits, which yield higher absolute RMSEs but are standard in our benchmark families (MARCEL, prior DKO work). On FreeSolv, D-MPNN achieves ~ 1.1 RMSE [14] vs. our SchNet’s 2.324. On QM9, equivariant GNNs achieve HOMO MAE ~ 15 meV [22] vs. our SchNet’s ~ 19 meV RMSE (different metric). These gaps reflect our focus on comparing representation strategies under controlled conditions (same splits, same conformers, same training protocol) rather than maximizing absolute performance. We do not benchmark D-MPNN [14, 41], Uni-Mol [43], or pre-trained models [32] because they address an orthogonal question (graph architecture and pre-training) to ours (when does 3D conformer geometry complement 2D descriptors).

Limitations of the taxonomy. Our property taxonomy is derived from a limited set of datasets (3 solvation, 4 steric, 7 electronic) and should be treated as a working hypothesis. Properties with mixed solvation/electronic character (e.g., membrane permeability) may not fit neatly into a single category. Learning curves (Section 4.10) reveal dataset size effects: FreeSolv hybrid improvement only emerges with >440 training molecules, suggesting the taxonomy should be conditioned on data availability. Validation on additional property types is needed before it can serve as a reliable practitioner guide.

The simplicity principle. Simpler representations consistently outperform complex ones. Complex covariance representations (low-rank, cross-attention) underperform simple ones (5 summary statistics, gating). With $D = 1024$, the full Σ has $\sim 500K$ unique entries. On datasets of 642–4,200 molecules, the additional parameters in richer representations overfit—a classic bias-variance trade-off where the bias of a compressed representation is preferable to the variance of an overparameterized one.

Practical implications. For practitioners, our results suggest: (1) always start with Morgan FP + XGBoost as a baseline, (2) if the property involves solvation or flexibility and the dataset contains >500 molecules, add conformer μ and Σ invariants—expect the largest gains for large, flexible molecules with multiple rotatable bonds, (3) consider enhanced physicochemical 3D descriptors (PMI, SASA, USR) which can outperform raw geometric features, and (4) for maximum accuracy, end-to-end 3D GNNs (SchNet) provide the strongest results. The scaffold split validation ensures these recommendations generalize to novel chemical scaffolds, not just structurally similar compounds.

6 Conclusion

We presented a systematic study of when 3D conformer geometry complements 2D molecular fingerprints, yielding three main contributions: (1) an empirical property taxonomy showing that conformer statistics selectively help solvation-dependent properties but not electronic or steric tasks, validated across 14 regression targets with 10-seed significance testing and scaffold splits; (2) a performance hierarchy—SchNet $\approx$ FP+3D $>$ FP+XGBoost $>$ neural conformer ensembles—revealing that the pre-computed feature bottleneck, not the absence of 3D information, limits conformer ensemble methods; and (3) mechanistic understanding of why fingerprints dominate (breadth of chemical information) and when conformer statistics help (solvation properties of large, flexible molecules).

For practitioners, the decision framework is: start with Morgan FP + XGBoost; add conformer μ statistics for solvation/flexibility properties with >500 molecules; consider physicochemical 3D descriptors (PMI, SASA, USR) for further gains; and use end-to-end 3D GNNs when maximum accuracy justifies the cost. Five to ten conformers suffice for hybrid models.

Limitations. Our conformer features are derived from RDKit ETKDG conformers, which may not accurately represent the true Boltzmann ensemble; the BDE pipeline uses uniform rather than energy-derived weights. While we validate key results under scaffold splits, the main benchmark uses random splits following prior work conventions. The hybrid approach uses simple feature concatenation; learned feature interaction might yield larger improvements. Error analysis stratification uses a single seed, so individual quartile improvements (e.g., +39.7% for rotatable bonds Q3) should be interpreted cautiously despite robust overall trends. Our SchNet baseline uses a single architecture; future work should

explore PaiNN [36], DimeNet++ [8], and other equivariant GNNs. Larger-scale validation on ChEMBL/ZINC datasets and Boltzmann-weighted conformer generation remain important directions.

Reproducibility. Code and data will be released upon acceptance.

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A Full Per-Dataset Results

Table 14 presents the complete benchmark results for ESOL and FreeSolv with all Phase 1 models, including baseline architectures. Models are described in Section 3.3: **single_conformer** uses only the lowest-energy conformer; **DeepSets** [42] applies permutation-invariant aggregation; **dko_separate_nets** uses independent μ/Σ networks without kernel fusion. Full results for all 6 regression datasets are available in the supplementary materials.

Table 14: Full results on ESOL and FreeSolv (RMSE $\pm$ std, 3 seeds).

Model	ESOL	FreeSolv
dko_first_order	1.646 $\pm$ 0.036	4.513 $\pm$ 0.110
attention	1.888 $\pm$ 0.011	4.077 $\pm$ 0.071
mean_ensemble	2.016 $\pm$ 0.070	4.177 $\pm$ 0.048
dko_diagonal	2.040 $\pm$ 0.033	4.699 $\pm$ 0.057
dko	2.056 $\pm$ 0.030	4.881 $\pm$ 0.140
dko_separate_nets	2.249 $\pm$ 0.031	4.790 $\pm$ 0.134
single_conformer	2.394 $\pm$ 0.120	4.087 $\pm$ 0.100
DeepSets	2.463 $\pm$ 0.637	4.277 $\pm$ 0.177

B Implementation Sensitivity Analysis

The DKO implementation is sensitive to several hyperparameters, identified through systematic ablation:

- (1) **Learning rate** (10^{-5} vs 10^{-4}): Using a $10\times$ lower learning rate than other models reduced ESOL RMSE by 30% ($3.318 \rightarrow 2.309$) when corrected.
- (2) **Feature normalization** ($\dim=(1, 2)$ vs $\dim=1$): Normalizing across both conformers and features destroyed inter-feature variance in Σ , making all covariance features approximately identical. Correcting to $\dim=1$ restored Pearson correlation from ~ 0 to ~ 0.48 .
- (3) **Kernel output dimension** (32 vs 64): Marginal individual effect but needed for combined benefit.
- (4) **Sigma regularization** (10^{-4} vs 10^{-2}): Too-small regularization caused near-singular covariance matrices.

- (5) **Mixed precision**: FP16 arithmetic is insufficient for covariance matrix operations. Enabling mixed precision increased RMSE by 30% with $10\times$ higher variance.

Combined, these corrections improved DKO’s ESOL R^2 from -1.70 to -0.04 , a swing of 1.66 in explained variance. The sensitivity of covariance-based methods to numerical precision and normalization choices is an important practical consideration.

C Conformer Diversity Analysis

Table 15 reports conformer diversity metric (CDM) statistics per dataset. CDM values are specific to our feature encoding and not directly transferable to other pipelines.

Table 15: Conformer diversity metric (CDM = $\text{tr}(\Sigma)$) statistics by dataset. Values are feature-encoding-specific.

Dataset	Median	Max	Q1	Q3
FreeSolv	0.01	42.8	0.0	24.95
ESOL	27.88	148.04	0.0	67.32
QM9	25.05	113.76	13.73	49.08
Lipophilicity	69.99	155.53	34.80	98.83

Lipophilicity has $3\times$ higher median CDM than QM9, consistent with dko_invariants achieving its strongest results on this dataset. FreeSolv has near-zero median CDM yet still benefits from hybrid features—primarily through first-order (μ) statistics (Table 5), confirming that CDM does not predict first-order conformer utility.

D Enhanced 3D Descriptors Benchmark

Table 16 presents the full benchmark of enhanced 3D descriptors across all feature combinations and datasets (mean $\pm$ std, 3 seeds). The 28 descriptors span four categories: **Shape** (6): PMI ratios, asphericity, eccentricity, radius of gyration, inertial shape factor; **Surface** (5): SASA total, polar, nonpolar, and ratios; **USR** (12): distances from centroid, closest/farthest atom references; **Global** (5): span, bounding box, volume, compactness.

Table 16: Full 3D descriptor benchmark (RMSE, mean of 3 seeds). “3D” = 28 physicochemical descriptors; “Geo” = geometric $\mu+\Sigma$ features.

Features	ESOL	FreeSolv	Lipo	QM9-Gap
3D-only (μ)	1.340	4.053	1.119	0.035
3D $\mu+\sigma$	1.359	4.100	1.094	0.034
FP only	1.507	2.939	0.910	0.020
FP + 3D μ	1.025	2.936	0.868	0.020
FP + 3D $\mu+\sigma$	1.000	3.073	0.873	0.020
FP + Geo $\mu+\sigma$	1.358	2.824	0.957	0.021
FP + 3D + Geo	1.094	2.597	0.916	0.020

Key observations: (i) 3D descriptors alone ($R^2 \approx 0.55$ on ESOL) outperform raw geometric μ features, validating

the physicochemical relevance of PMI/SASA/USR; (ii) combining FP+3D achieves the best non-GNN result on ESOL (1.000, $R^2 = 0.75$); (iii) adding geometric features to FP+3D helps on FreeSolv (2.597 vs. 3.073) but not ESOL (1.094 vs. 1.000), suggesting dataset-specific feature complementarity.

E Conformer Count Ablation

Table 17 shows the effect of conformer count on ESOL for both XGBoost (hybrid) and dko_gated (neural) models.

Table 17: Conformer count ablation on ESOL (mean of 3 seeds). XGBoost saturates at $n=5-10$; dko_gated improves monotonically.

n	XGB RMSE	XGB R^2	DKO RMSE	DKO R^2
1	1.401	0.519	5.487	<0
5	1.337	0.562	1.773	0.229
10	1.330	0.567	1.723	0.272
20	1.354	0.550	1.704	0.288
50	1.358	0.548	1.642	0.339

XGBoost reaches near-optimal performance at $n=5-10$ conformers (RMSE 1.330–1.337) with slight degradation at higher counts, likely due to noise in conformer mean statistics. In contrast, dko_gated improves monotonically from degenerate ($n=1$, $R^2 < 0$) to $R^2 = 0.34$ at $n=50$, suggesting the neural architecture benefits from richer distributional estimates. Practical implication: for XGBoost-based hybrid models, generating 5–10 conformers is sufficient, saving significant computational cost vs. the default $n=50$.

F SchNet Architecture Details

We use SchNet [35] with 128 hidden channels, 6 interaction blocks, 50 radial basis function (RBF) Gaussians, and a 10Å cutoff radius. Each molecule is represented by up to 10 conformers (lowest-energy from ETKDG); predictions are averaged across conformers. Training uses Adam [17] (not AdamW, as SchNet’s original implementation uses Adam) with learning rate 5×10^{-4} , batch size 32, and early stopping (patience 50). Per-seed results are shown in Table 18.

Table 18: SchNet per-seed results (RMSE / R^2 , 3 seeds).

Seed	ESOL	FreeSolv	Lipo
42	1.017 / 0.747	2.591 / 0.517	0.721 / 0.611
123	0.943 / 0.782	2.055 / 0.696	0.714 / 0.618
456	1.054 / 0.728	2.327 / 0.611	0.713 / 0.620
Mean	1.004 / 0.752	2.324 / 0.608	0.716 / 0.616

FreeSolv shows the highest variance across seeds (std = 0.268), consistent with its small dataset size (642 molecules). SchNet’s strong performance despite averaging over only 10 conformers (not ensemble statistics) suggests that end-to-end 3D learning captures geometric information more effectively than pre-computed summary statistics.

G Hybrid Neural MLP vs XGBoost

We compare a neural MLP (2309→512→256→128→1, ReLU activations, dropout 0.1) against XGBoost on the same hybrid features (FP + μ + Σ).

Table 19: Hybrid MLP vs. XGBoost (RMSE / R^2 , mean of 3 seeds).

Method	ESOL	FreeSolv	Lipo	QM9-Gap
MLP	1.218 / 0.637	3.809 / -0.044	0.906 / 0.394	0.023 / 0.787
XGBoost	1.358 / 0.548	2.824 / 0.423	0.957 / 0.325	0.021 / 0.821

MLP outperforms XGBoost on ESOL (1.218 vs. 1.358) and Lipophilicity (0.906 vs. 0.957), but XGBoost wins on FreeSolv (2.824 vs. 3.809) and QM9-Gap (0.021 vs. 0.023). The MLP’s negative R^2 on FreeSolv indicates overfitting on this small dataset (642 molecules), while XGBoost’s built-in regularization prevents this failure mode. This suggests that the optimal learning algorithm for hybrid features is dataset-dependent.

H Mutual Information Analysis

We estimate mutual information (MI) using the Kraskov–Stögbauer–Grassberger (KSG) k -nearest-neighbor estimator [19] with $k=5$, selecting the top 50 features by individual MI for each feature group. For MI estimation, we project the 1024-dimensional μ to 256 dimensions via PCA to mitigate the curse of dimensionality in KSG k -nearest-neighbor estimation.

Table 20: Mutual information analysis (nats, top-50 features per group). “Per-feat.” = mean MI per feature.

Dataset	Group	Total MI	Per-feat.
ESOL	FP (2048-bit)	1.10	0.022
	μ (256-dim PCA)	8.27	0.165
	Σ (5-dim)	0.35	0.070
FreeSolv	FP	2.28	0.046
	μ	4.39	0.088
	Σ	0.22	0.045
Lipo	FP	0.38	0.008
	μ	0.75	0.015
	Σ	0.03	0.006
QM9-Gap	FP	1.79	0.036
	μ	9.70	0.194
	Σ	0.54	0.108

Key findings: (i) μ features carry 2–8× more MI per feature than FP bits across all datasets; (ii) despite higher per-feature informativeness, μ ’s advantage depends on dataset—ESOL shows 7.5× ratio while FreeSolv shows only 1.9×; (iii) Σ features have moderate per-feature MI (0.006–0.108 nats) but contribute minimal *conditional* MI given FP+ μ (−1.5% to −3.5%, Table 21).

Table 21: Conditional MI of Σ given FP+ μ (nats). Negative values indicate Σ is redundant.

Dataset	MI(FP+ μ)	MI(FP+ μ + Σ)	$\Delta\%$
ESOL	8.27	7.98	-3.5%
FreeSolv	4.39	4.32	-1.5%
Lipo	0.75	0.74	-2.2%
QM9-Gap	9.70	9.39	-3.2%

Limitations: KSG MI estimation is sensitive to k and dimensionality. The top-50 feature selection introduces bias toward groups with more features (μ : 256 candidates, FP: 2048 candidates, Σ : 5 candidates). Conditional MI estimates may be noisy for high-dimensional feature spaces.

I Feature Attribution

We report XGBoost feature importance (gain-based) aggregated by feature group across 3 seeds.

Table 22: Feature group importance (% of total gain). Σ contributes <2% across all datasets.

Dataset	FP (%)	μ (%)	Σ (%)
ESOL	31.4	67.4	1.2
FreeSolv	62.0	36.7	1.3
QM9-Gap	49.3	49.0	1.7

Top-10 features by dataset. On ESOL, 7 of the top 10 features are μ dimensions (mu_dim_137, mu_dim_138, mu_dim_131 are top 3), reflecting that mean conformer geometry is the primary signal for solubility. On FreeSolv, all top 10 are fingerprint bits (FP_bit_314, FP_bit_807, FP_bit_1114), with μ features first appearing at rank 15. On QM9-Gap, the top 10 alternate between μ (rank 1: mu_dim_146) and FP (rank 2: FP_bit_1380), reflecting the balanced importance.

Among the 5 Σ features, top5_var and effective_rank contribute the most across datasets, while total_var and max_var are generally less important. The sigma feature sigma_mean_var appears at rank 17 on QM9-Gap (importance 0.85%), the highest-ranked Σ feature on any dataset. Note that the hybrid XGBoost model uses variance-based Σ features (σ_5 ; Section 3.5), while dko_invariants uses matrix-invariant features (Section 3.3).

J Scaffold Split Validation Details

Table 23 presents the full scaffold split results including Mu-only features.

Scaffold splits are substantially harder than random splits across all datasets and feature configurations, with FP-only RMSE increasing by 38% (ESOL), 83% (FreeSolv), and 4% (Lipophilicity). The small Lipophilicity degradation suggests that octanol–water partition coefficients generalize well across scaffolds, consistent with this property being primarily determined by local functional group contributions. Mu-only features consistently underperform FP-only under both split

Table 23: Full scaffold vs. random split results (RMSE, mean $\pm$ std, 3 seeds).

Dataset	Split	FP-only	FP+ μ + Σ	μ -only
ESOL	random	1.083 $\pm$ 0.057	0.991 $\pm$ 0.083	1.315 $\pm$ 0.056
	scaffold	1.492 $\pm$ 0.016	1.315 $\pm$ 0.019	1.595 $\pm$ 0.002
FreeSolv	random	1.701 $\pm$ 0.093	2.008 $\pm$ 0.199	3.011 $\pm$ 0.261
	scaffold	3.107 $\pm$ 0.114	3.173 $\pm$ 0.154	3.935 $\pm$ 0.147
Lipo	random	0.881 $\pm$ 0.028	0.887 $\pm$ 0.039	1.099 $\pm$ 0.033
	scaffold	0.914 $\pm$ 0.001	0.937 $\pm$ 0.006	1.097 $\pm$ 0.012

types, confirming that conformer features alone cannot compete with fingerprints for generalization.

K Learning Curve Details

Table 24 presents the full learning curve results with absolute RMSE values.

Table 24: Learning curves: absolute RMSE (mean of 3 seeds) at each training fraction.

Frac	ESOL		FreeSolv		Lipo		QM9-Gap	
	FP	Hyb	FP	Hyb	FP	Hyb	FP	Hyb
10%	2.007	1.913	3.597	4.096	1.066	1.131	.0241	.0279
25%	1.744	1.665	3.135	3.592	0.966	1.056	.0219	.0244
50%	1.597	1.521	3.221	3.328	0.950	0.985	.0209	.0224
75%	1.578	1.419	3.056	2.988	0.928	0.956	.0205	.0215
100%	1.533	1.343	3.077	2.862	0.920	0.932	.0201	.0208

The opposite scaling patterns across datasets are striking: on ESOL and FreeSolv, hybrid features become increasingly valuable with more data (suggesting genuine signal), while on Lipophilicity and QM9-Gap, the penalty diminishes (suggesting XGBoost learns to downweight irrelevant features). The FreeSolv crossover at $\sim$ 75% ($\sim$ 440 molecules) provides a practical threshold for when conformer features become beneficial on small solvation datasets.

L Error Analysis Details

FreeSolv (overall: -2.8%): Larger molecules benefit from hybrid features (Q4 heavy atoms: +7.0%, Q4 molecular weight: +7.4%) while small molecules are hurt (Q1 molecular weight: -34.5%). Molecular weight significantly correlates with hybrid improvement (Spearman $r = 0.30$, $p = 0.025$), as does heavy atom count ($r = 0.28$, $p = 0.042$).

Lipophilicity (overall: -5.7%): Hybrid features hurt uniformly across all quartiles of all molecular properties. No significant correlations between any molecular descriptor and hybrid improvement (all $p > 0.29$). This confirms that conformer features are genuinely uninformative for lipophilicity prediction, regardless of molecular size, flexibility, or surface properties.

ESOL TPSA analysis: Improvement is relatively uniform across TPSA quartiles (+3.9% to +13.6%), suggesting

that hybrid benefit is not driven by polar surface area. The strongest signals are molecular size (heavy atoms, MW) and

flexibility (rotatable bonds), both of which directly determine conformer diversity.